

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 902+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

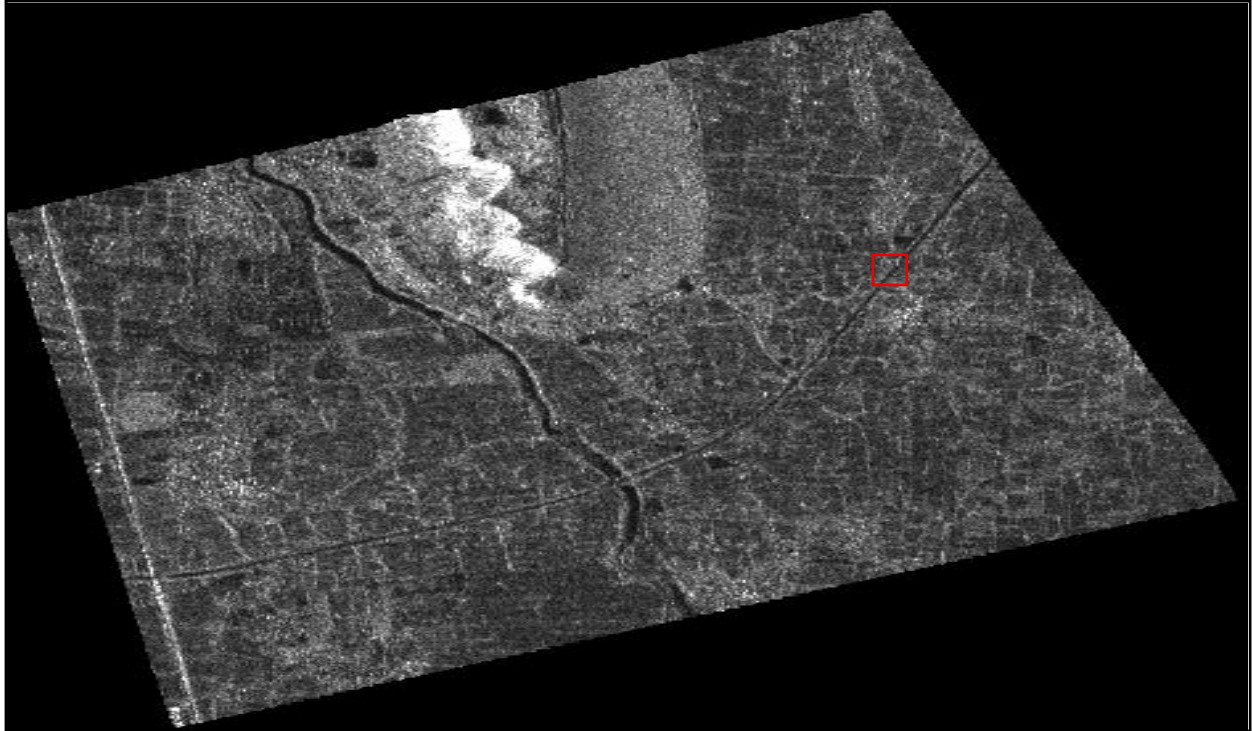
Flight Direction - DESCENDING

Polarization – VV+VH

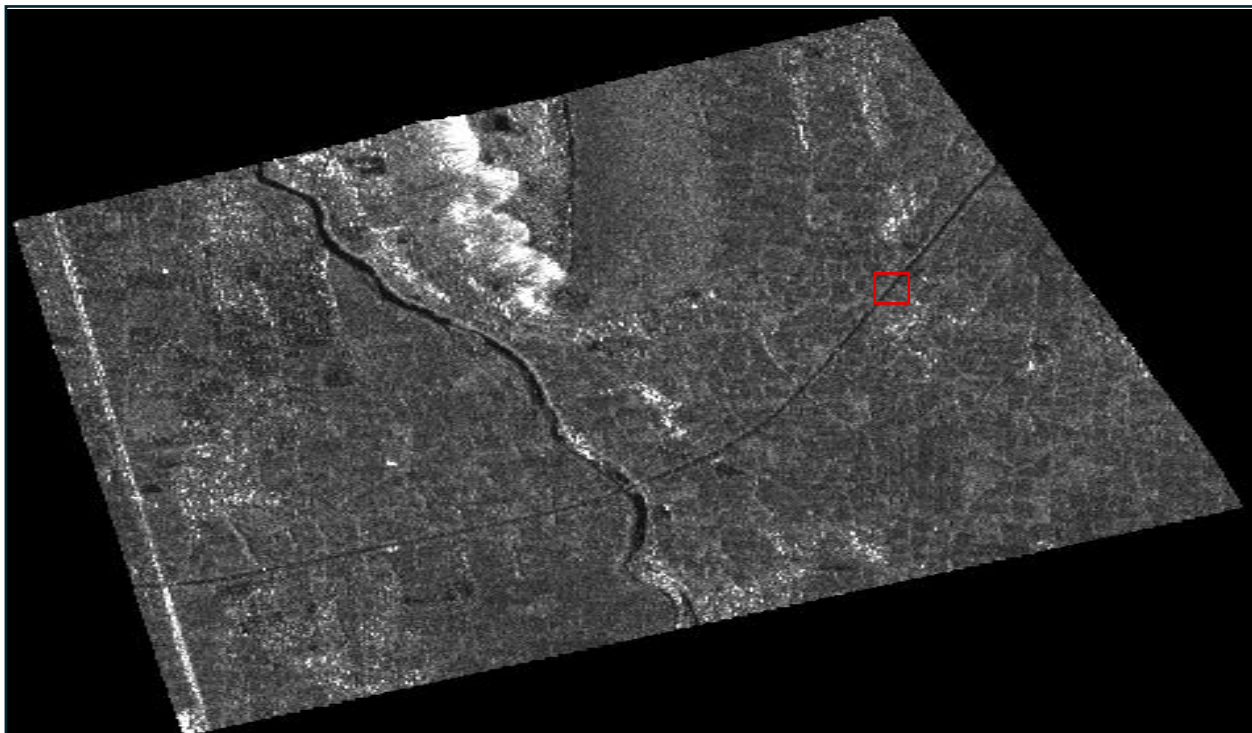
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

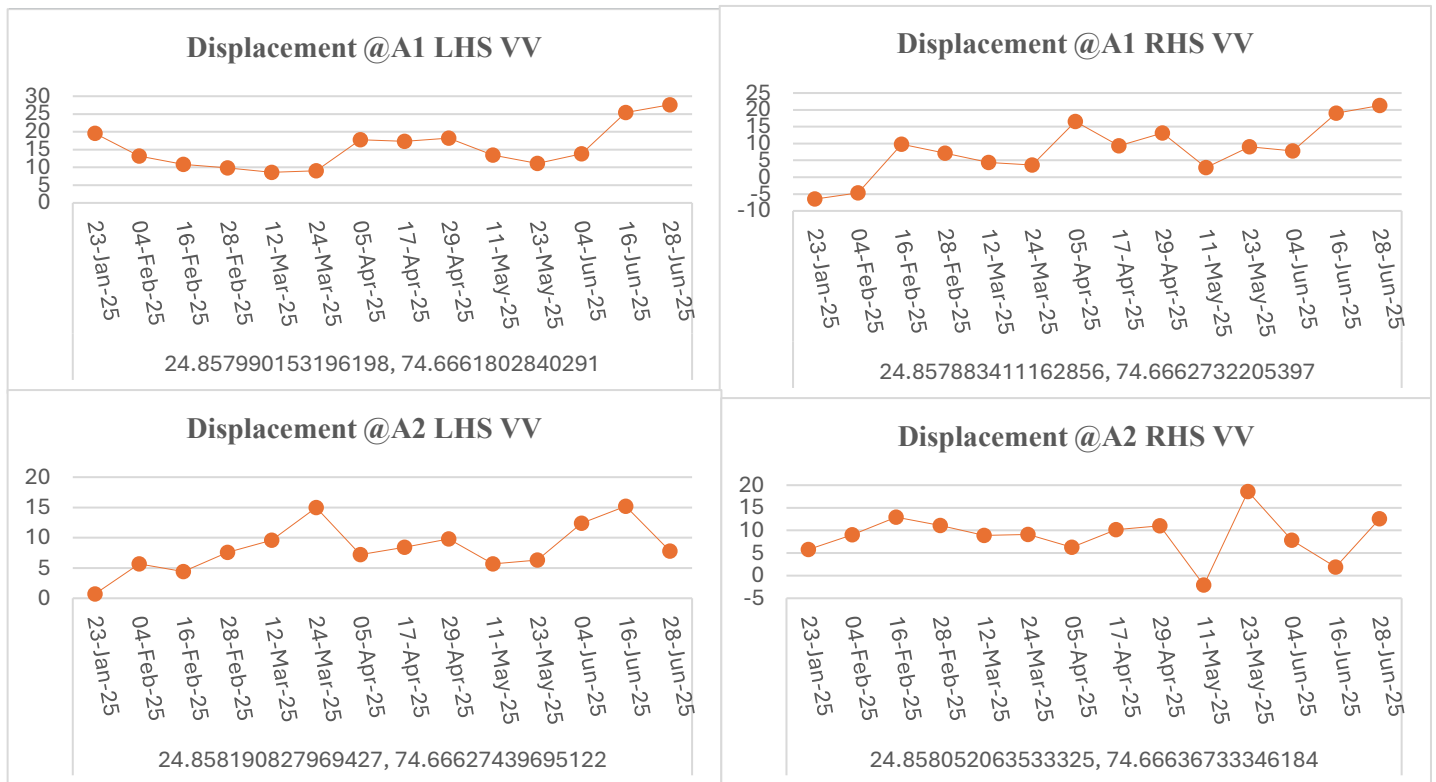
VUP – CH-902+200

Length – 20.78M

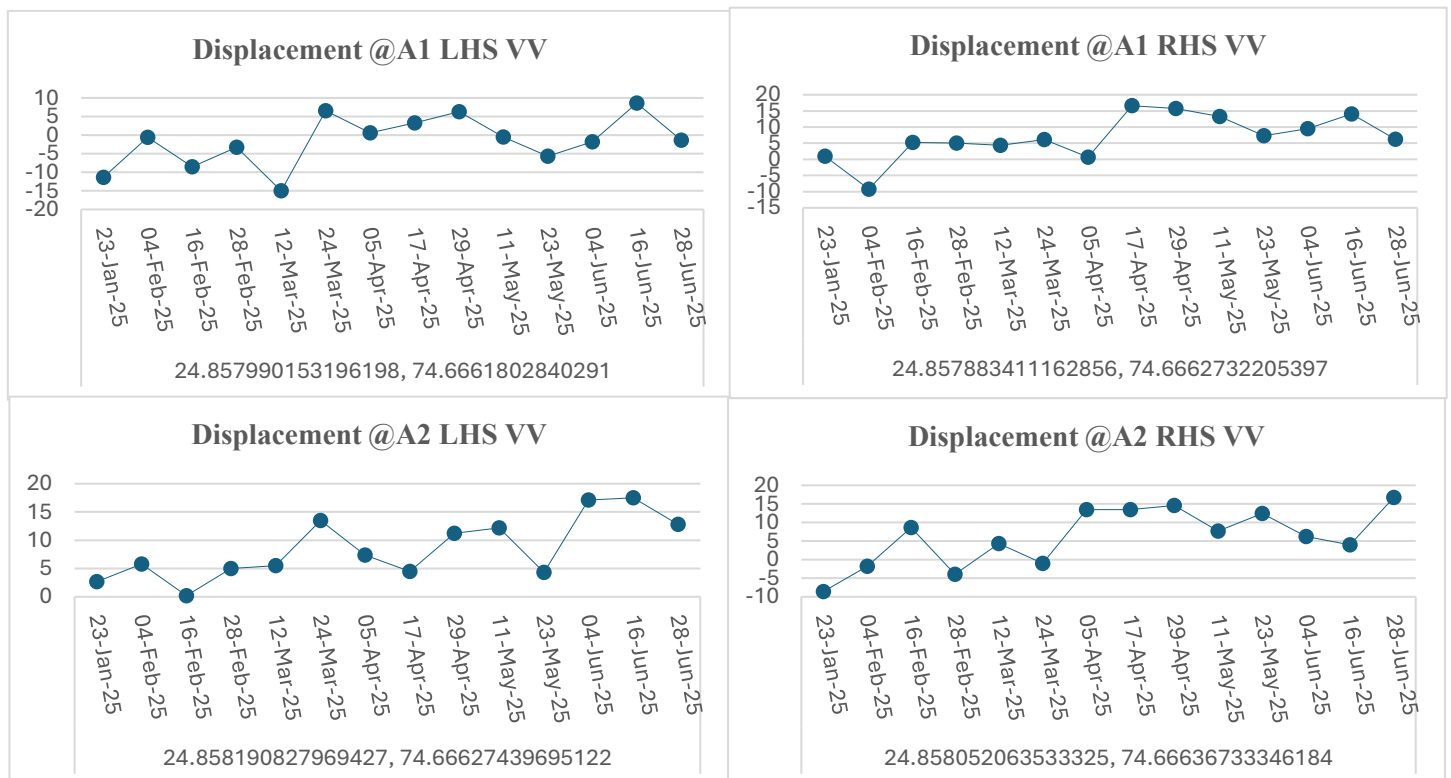


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



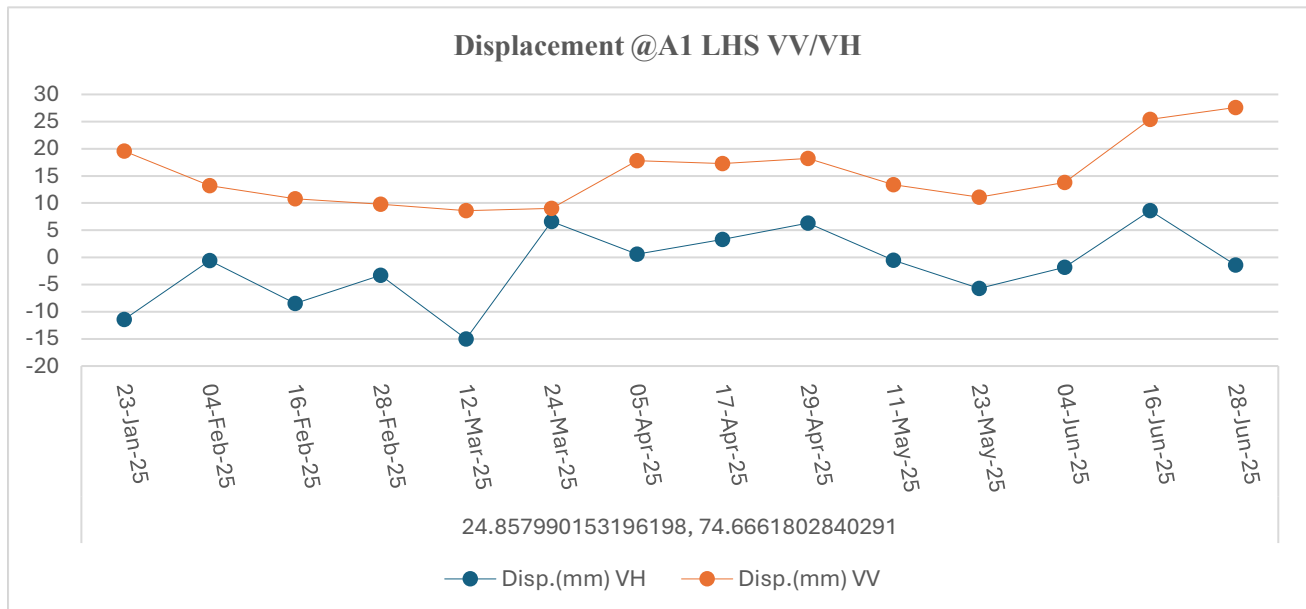
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.857990153196198, 74.6661802840291)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement varies from approximately -4 mm to +16 mm, showing moderate vertical variability with alternating uplift and settlement phases.
- **Horizontal Displacement (VH)** Horizontal displacement at A2 RHS ranges approximately from +6 mm to -22 mm. The response is predominantly negative.

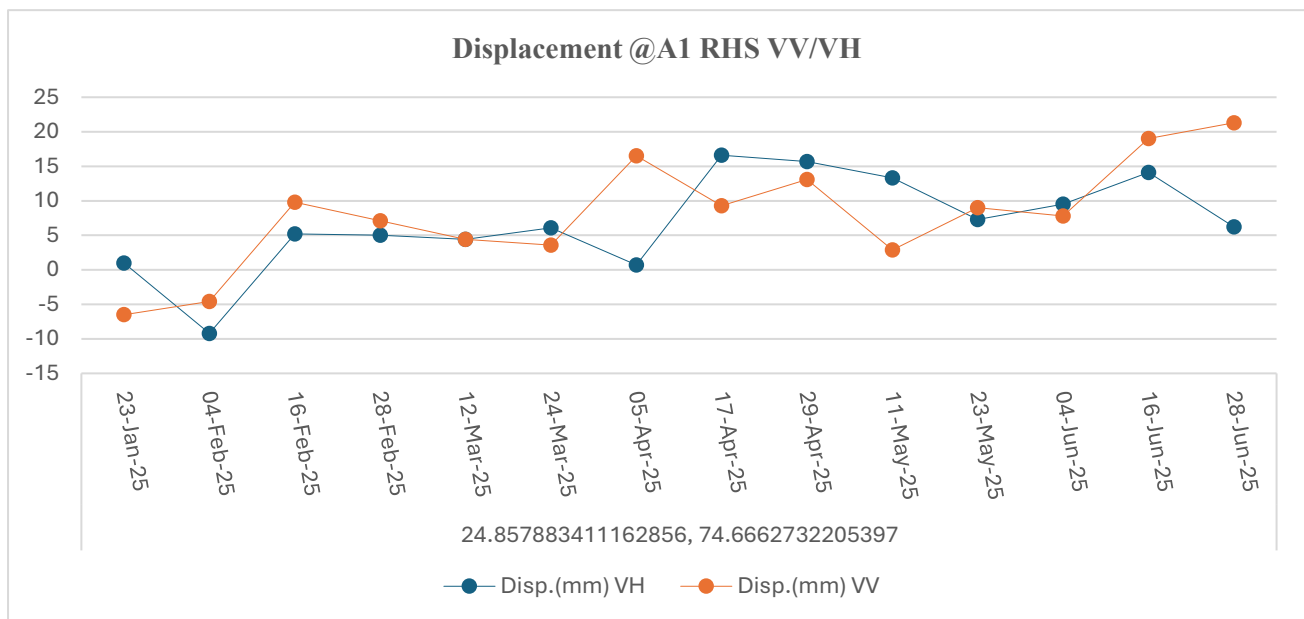
Critical Events & Anomalies:

- **February 2025:** VH briefly shifts into the positive range while VV rises to about +10 mm.
- **May 2025:** VH remains consistently negative (around -12 to -15 mm), while VV increases toward +15 mm, suggesting continued lateral displacement with vertical recovery or uplift
- **June 2025:** VH records another major negative peak as VV attains its maximum positive value indicating significant lateral movement occurring simultaneously with vertical uplift, a critical condition warranting close observation.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.857883411162856, 74.6662732205397)



Key Observations:

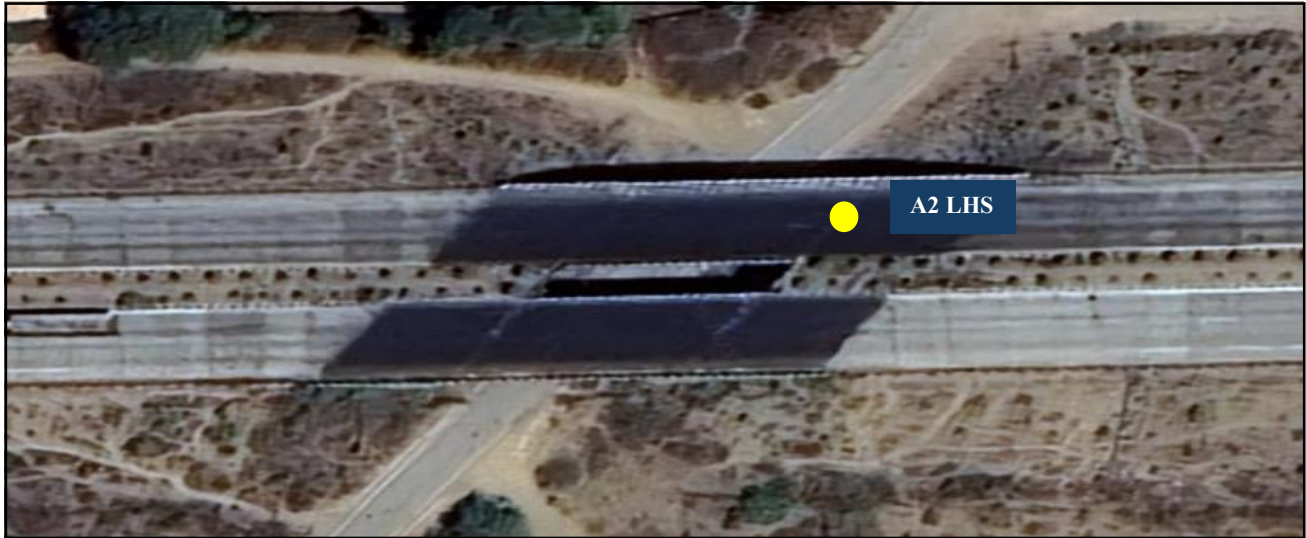
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement at A1 RHS ranges approximately from -9 mm to +16 mm. The response shows an overall positive bias, with short-term negative excursions early.
- **Vertical displacement (VV)** Vertical displacement varies from approximately -7 mm to +21 mm, indicating moderate to high vertical movement.

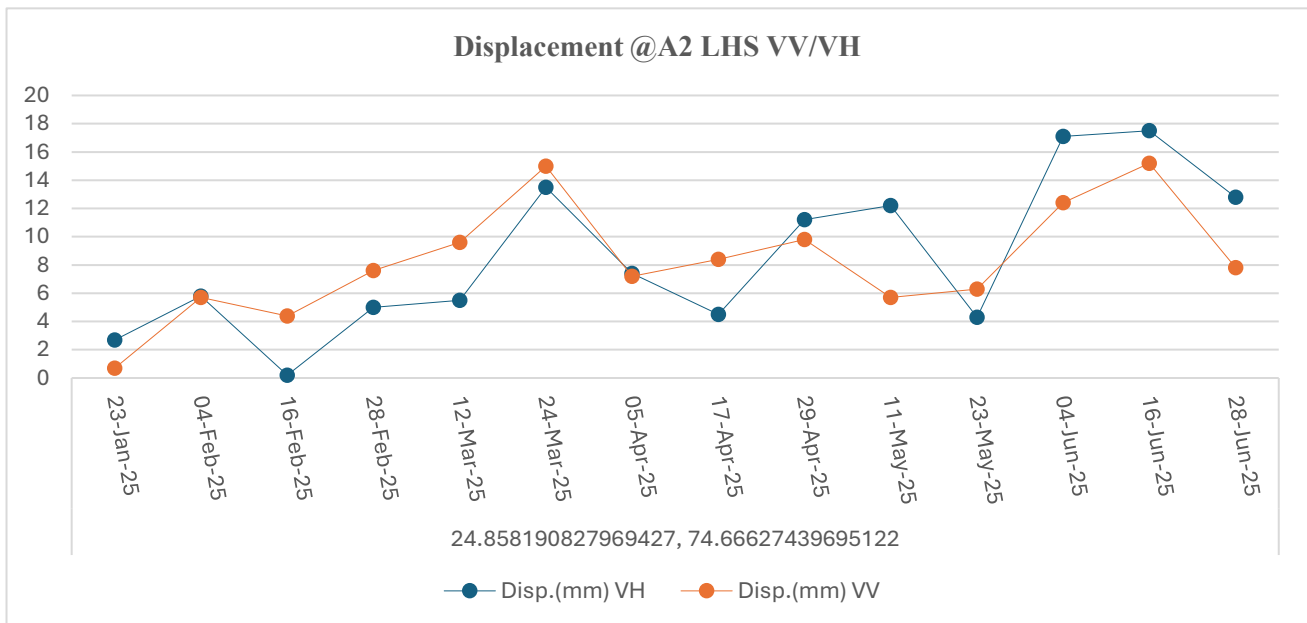
Critical Events & Anomalies:

- **February 2025:** VH reaches its minimum value while VV remains slightly negative, indicating an initial phase of combined lateral and vertical settlement.
- **April 2025:** VV records a sharp uplift peak while VH briefly drops close to zero, indicating a dominant vertical response with reduced lateral movement.
- **June 2025:** VV increases steadily to its maximum while VH remains positive but moderates toward +6 mm, suggesting continued vertical uplift with stabilized lateral displacement.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.858190827969427, 74.66627439695122)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** at A2 LHS varies approximately from 1 mm to 17 mm, remaining predominantly positive throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately from 2 mm to 15 mm, showing consistent uplift-dominated behaviour with moderate fluctuations and no prolonged settlement phase.

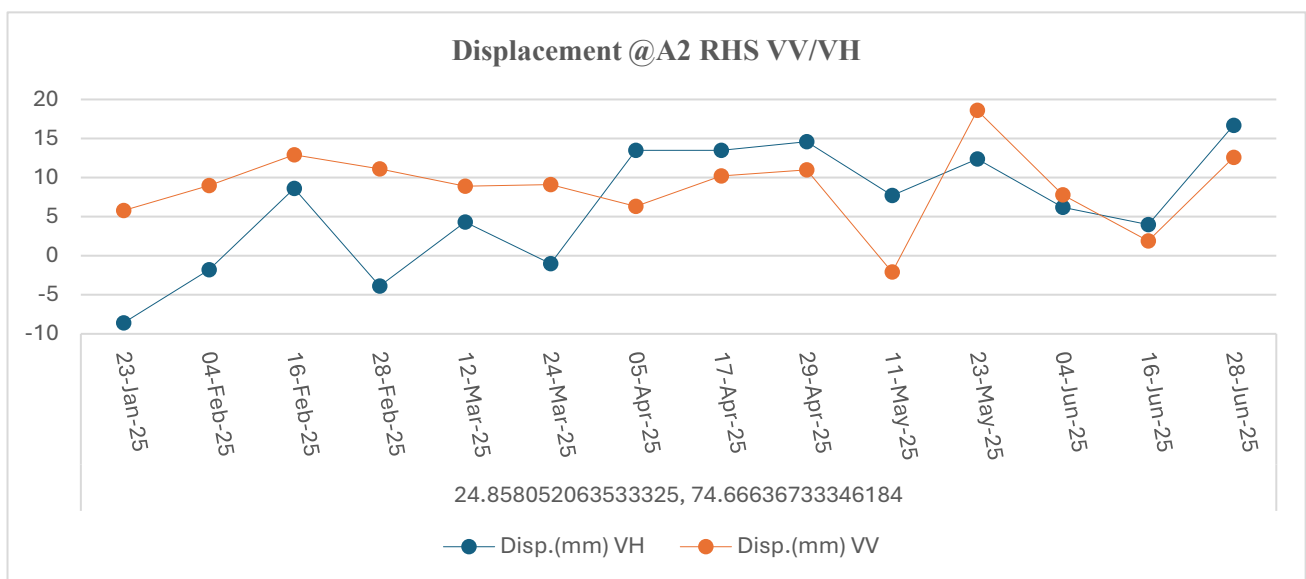
Critical Events & Anomalies:

- **February 2025:** Both VH and VV remain low and positive, indicating an initial stable structural condition with minor lateral and vertical movement.
- **June 2025:** VH reaches its maximum values alongside elevated VV, indicating sustained lateral displacement combined with continued uplift, suggestive of long-term deformation stabilization in a positive displacement regime.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.858052063533325, 74.66636733346184)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement exhibits moderate variability with a predominantly positive trend over the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacement shows high variability with an overall positive dominance, ranging roughly between -9 mm and +17 mm.

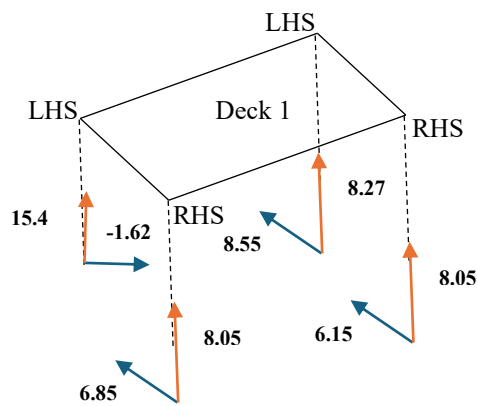
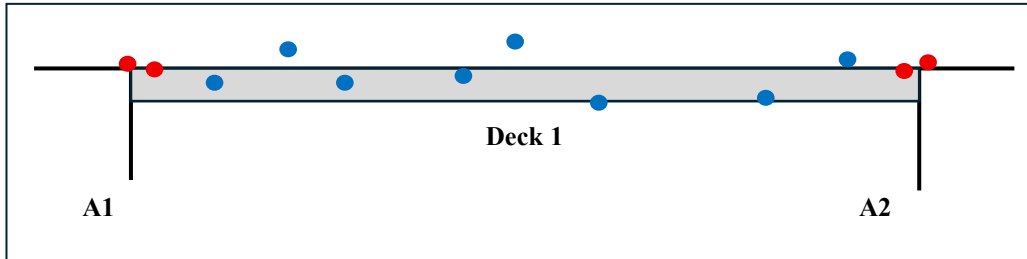
Critical Events & Anomalies:

- **February 2025:** VH drops to -4 mm, while VV remains elevated, indicating a decoupled response with temporary lateral relaxation.
- **March 2025:** VH dips slightly to -1 mm, while VV remains positive, indicating minor lateral softening without vertical deterioration.
- **June 2025:** VH rises to +16 mm, while VV recovers to +13 mm, confirming renewed combined lateral and vertical uplift toward the end of the monitoring period.

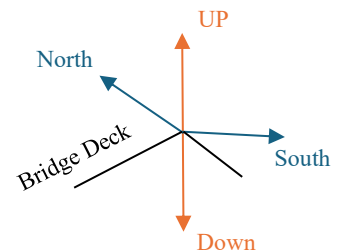
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits predominantly upward vertical displacement, indicating an overall uplift response across the deck. The maximum vertical uplift is observed at the LHS upper deck region, reaching approximately +15.4 mm, representing the most significant vertical response in this case.

Horizontal Displacement Behaviour:

- Horizontal displacement values indicate moderate lateral movement with directional variation, suggesting torsional and bending interaction within the deck slab. A minor opposing horizontal displacement (−1.6 mm) is observed near the upper LHS region, suggesting localized restraint or stiffness variation at that support location.

CONCLUSION:

- The bridge deck slab demonstrates a predominantly uplift-dominated vertical response, contrasting with previously observed settlement-dominated cases. Overall, the displacement magnitudes remain within a moderate response range, though the LHS upper region emerges as a critical monitoring location due to peak vertical movement.